A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 16

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** B (about 40:40:20 sand:silt:clay) **1** Balanced mix gives good drainage and aeration from the sand fraction **1** Water retention and nutrient (ion) retention from the clay fraction – easy root penetration and cultivation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **loam** → a balanced mixture of sand, silt and clay	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) **1** Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand **1** With only 5% clay there is little surface to hold ions so nutrients are leached – low water retention too **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeability** → how easily water passes through / lets water through • **adsorb** → nutrient ions sticking to the surfaces of clay and humus • **leached** → nutrients or ions washed down out of the soil / washed away by water	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** C (clay) holds much water so has a high thermal capacity: warms and cools slowly – A warms rapidly after cold weather **1** Clay particles held together by capillary water so roots find it hard to push between them – penetration is easy in sandy A **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Shake a known mass of soil with water in a measuring cylinder and leave to settle – particles settle in order of size (sand first, then silt, then clay; organic matter floats) **1** Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) **1** Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (930 - 250) / 250 x 100 **1** = 272% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.3	**1 mark each** 1 mark for each of the 3 points below. **1 Anticyclonic conditions:** clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled **1** Low wind velocity so air layers did not mix – London lies in a valley / basin where cold dense air collected: a temperature inversion formed **1** Smoke from coal fires was trapped below the inversion – moist air cooled to its dew point forming fog – smoke + fog = smog – the smog's high albedo prolonged the inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **temperature inversion** → a layer of warm air above cooler air that traps pollutants • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Smoke particles (PM10) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections **1** These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust	2
02.5	1 mark for the point below. **1** Clean Air Act (1956) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) **1** Radon travels up cracks in the rock and through foundations into buildings **1** It becomes concentrated inside enclosed / poorly ventilated buildings – control by ventilating foundations / sumps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16 + 1.5 = 17.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
03.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Where they live (geology / radon, altitude) **1** Occupation (radiographer, nuclear worker, aircrew, miner) **1** Medical history (number of X-rays / CT scans) **1** Lifestyle / diet (eg shellfish near discharges) – smoking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it – risks can be reduced (shielding, dose limits) **1** Symptoms of chronic low-dose exposure take a long time to develop **1** The same symptoms (cancers) have other causes so effects are hard to attribute **1** Accurate data on human exposure / doses is often unavailable (especially after accidents) **1** People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **Accurate** → close to the true value	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	
04.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) **1** Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles **1 Carbon neutral in principle:** CO2 released equals CO2 absorbed in photosynthesis **1** Energy density of alcohol / biodiesel nearly as high as fossil fuels – wastes (straw, sewage, landfill gas) are cheap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **landfill gas** → the methane and carbon dioxide produced by anaerobic decomposition of waste **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) **1** Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO2 as using fossil fuels directly **1** Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally **1** Supply from wastes limited to the amount produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
04.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Hydrogen from algae:** some algae deprived of sulfur produce hydrogen during photosynthesis **1** Anaerobic digestion extended to crop wastes and marine algae – small-scale digesters (eg one cow's manure supplying a family's cooking gas) **1** Biofuels from microorganisms (eg oil-producing microalgae) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK **1** New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** A: Student's t-test - a difference between the means of two sets of normally distributed data **1** B: chi-squared test - a comparison of observed frequencies (counts) with expected frequencies **1** C: Spearman's rank correlation - a relationship between two variables measured on the same sampling points **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Student's t** → a test for a difference between the means of two normally distributed sets • **chi-squared** → a test comparing observed frequencies with expected frequencies • **Spearman** → a test for correlation between two sets of ranked data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 2% probability, or a 1 in 50 chance, that a result at least as extreme as this would arise by chance alone if the null hypothesis were true **1** This is below the 5% significance level normally used, so the null hypothesis is rejected and the difference or correlation is described as statistically significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **significance level** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** The calculated value on its own has no meaning: it must be compared with the value that would be exceeded by chance in 5% of cases, which is read from a table of critical values **1** The critical value depends on the number of degrees of freedom, which comes from the number of categories or pairs of data, because a larger sample makes a given result less likely to be chance **1** If the calculated value exceeds the critical value at p = 0.05 the null hypothesis is rejected – if it does not, the null hypothesis is not rejected and the result is not significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **critical value** → the table value the calculated statistic must reach to be significant • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** The standard deviation measures the spread of the individual values about the mean, so a small value means the heights were closely grouped and the mean is a good summary of them **1** Comparing the standard deviations with the difference between the means shows whether the two sets really are distinct or merely overlap **1** It is drawn as error bars above and below each plotted mean – where the error bars of two means do not overlap, a significant difference is likely, and where they overlap widely it is not **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	3
06.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** A correlation shows only that the two variables change together – it suggests a link that further work would have to test **1** A third factor may cause both, for instance shade, which raises soil moisture by reducing evaporation and also directly favours the moss **1** The relationship may run the other way round: a dense moss cover holds water and shades the soil, so the moss could be raising the moisture rather than responding to it **1** The points along one transect are close together and not independent, so a pattern may reflect one gradient across the site rather than a general relationship **1** Only an experiment in which soil moisture is changed while every other variable is held constant can establish cause **1** A result significant at p = 0.05 will still arise by chance about 1 time in 20, so a single significant result is not proof **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together • **transect** → a line along which samples are taken to show change with distance • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407, 420-423; AQA spec 3.7.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Molten iron from the blast furnace is converted directly to steel on the same site **1** So it does not cool and need re-melting – transport energy between sites is also avoided – waste heat from one process used by another (eg Kalundborg) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Collecting and processing small dispersed quantities can use more energy in transport than is saved **1** Mass production and large-scale mining have economies of scale – some materials (glass) take more energy to make than plastic but can be refilled many times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2
07.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared) – overhead cables cheaper than underground which need cooling **2** ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak	4
07.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Proximity to the grid / demand to reduce transmission losses and cable costs **1** Access to cooling water for thermal stations **1** Resource availability (wind, tidal range, fuel deliveries) **1** Land availability and planning / public acceptance – stable geology **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	
08.1	1 mark for the point below. **1** As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
08.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** About 39 °C (accept 38-40) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
08.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) **1** = 1.2 mg dm^{-3} (accept 1.1-1.3) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (mg dm^{-3}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Figure 1 shows the maximum / saturation values – processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line **1** Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster – values above the line are impossible **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
08.5	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Increased metabolic rate so more energy used / less surplus for growth **1** Eggs develop faster / hatch earlier (out of step with food supply) **1** Non-indigenous warm-water species may thrive and out-compete indigenous species **1** Toxicity of other pollutants is greater as toxins are metabolised faster **1** Reduced resistance to disease **1** Proteins / enzymes denatured if temperature is very high (above about 40 °C) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.6	1 mark for the point below. **1** Hot water is sprayed from a shower floor in a tall hollow tower – air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Control of greenhouse gases:** energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • **Geoengineering:** white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • **International agreements:** Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation – carbon trading and taxes – comparison of per capita carbon footprints between countries • **Evidence of limited success:** global CO2 concentrations still rising (about 280 ppm pre-industrial to over 410 ppm) – temperatures up about 1 °C – emissions in some developed countries falling but rising in developing economies • **Reasons for difficulty:** fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • **Judgement:** partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise – tipping points make urgency greater **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9
	• Any accurate, relevant material that is not in the indicative content. • **energy conservation** → using less energy, eg by insulation or behaviour change • **CCS** → capturing carbon dioxide from flue gases and storing it underground **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Strategies fall into control of greenhouse gases, geoengineering, international agreements and adaptation. Emissions are cut by energy conservation in transport, buildings and industry, by replacing fossil fuels with renewables and nuclear, by carbon capture and storage, by reducing deforestation and planting forests, and by capturing methane from landfill. Geoengineering ideas - white roofs, ocean fertilisation, solar shades - remain largely untried with unpredictable side effects. Adaptation, such as flood defences and managed retreat, reduces impacts but not the cause.	
	The Kyoto Protocol and Paris Agreement set targets, but they are weak or non-binding and participation is uneven; carbon trading and taxes exist but per capita footprints still differ enormously between countries. The evidence of success is limited: atmospheric CO_2 has risen from about 280 ppm before industrialisation to over 410 ppm and temperature by about 1 °C, and although emissions are falling in some developed countries they are rising in developing economies.	
	The difficulty is that fossil fuels underpin economies, so control demands lifestyle change, cost, and political agreement between nations with different priorities, while time delays and natural fluctuations mask the effect of any action. By contrast the Montreal Protocol succeeded because ozone was a substitution problem. The judgement must be partial success: the technology exists and regional reductions prove it works, but global strategies have not yet stopped the rise, and tipping points make the delay dangerous.	
	Weaker answer - Level 2 (5 marks)	
	Methods to reduce climate change include using renewable energy such as wind and solar instead of fossil fuels, saving energy, and planting trees to absorb carbon dioxide. International agreements like the Kyoto Protocol and the Paris Agreement have set targets for countries to reduce emissions. However carbon dioxide levels are still rising, so the methods have not been very successful so far. This is because countries rely on fossil fuels and do not always keep to the agreements. More needs to be done.	
	What makes the difference	
	• Level 2 covers the main strategies and reaches a conclusion, but with no quantitative evidence (280 to 410 ppm, 1 °C) and no mention of geoengineering or adaptation.	
	• Level 3 explains *why* success is limited (economic dependence, time lags, natural variability, political disagreement) rather than just stating that it is.	
	• The Montreal comparison and the tipping-point point give Level 3 the breadth and judgement the top band requires.	
10.2	1 mark for the point below. **1** The total greenhouse gas / CO2 emissions caused by an individual, organisation, product or country (often expressed per capita per year) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	
11.1	**Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Features of fertile soil:** water, nutrients, air, organic matter, pH, biota, texture, structure, depth – slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • **Causes of degradation:** vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • **Types of erosion:** rain splash, runoff, wind, slumping – consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • **Management:** long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy	25
	• Evaluation and judgement on which strategies are most effective in different environments *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Nature of the problem:** interconnected atmosphere-hydrosphere-biosphere systems – changes over different time and spatial scales – natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change – time delay between cause and effect (ocean heat capacity) • **Limitations of historic data:** sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • **Proxy data and their reliability:** dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • **Modern monitoring:** long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty – ecological uncertainty (species survival, distribution, fragmentation) • **Computer models:** how they are tested and refined – importance of accurate representative data – scenario-based predictions and ranges	25
	• Uncertainty over ENSO / Conveyor / jet stream responses • **Evaluation and conclusion:** confidence in the overall trend and human cause despite detailed uncertainty – role of peer review and international bodies (IPCC) *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	

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